

Name:- Vrushabh Sanjay Gite

PRN:- 2125UCEM2002

Assignment 3: Understanding Token Log Probabilities

Objective

.Learn how LLMs assign probabilities to generated tokens

Program

```
from openai import OpenAI
import math

# Replace with your actual API key
client = OpenAI(api_key="YOUR_API_KEY")

# Get user input
prompt = input("Enter your prompt: ")

# Generate response
response = client.responses.create(
    model="gpt-4.1",
    input=prompt,
    max_output_tokens=100,
    logprobs=True,
)

print("\n=====")
print("Generated Response")
print("=====")
print(response.output_text)

print("\n=====")
print("Token Log Probabilities")
print("=====")

print(f'{"Token":<20} {"Log Prob":<12} {"Probability":<15} {"Cumulative":<15} {"Confidence"}')
print("-" * 90)

cumulative_probability = 1.0

# Loop through each generated token
for token_data in response.output[0].content[0].logprobs:
    token = token_data.token
```

```

log_prob = token_data.logprob

probability = math.exp(log_prob)
cumulative_probability *= probability

# Confidence Color
if probability >= 0.80:
    confidence = " Green"
elif probability >= 0.50:
    confidence = " Yellow"
else:
    confidence = " Red"

print(
    f"{repr(token):<20} "
    f"{log_prob:<12.4f} "
    f"{probability:<15.6f} "
    f"{cumulative_probability:<15.6f} "
    f"{confidence}"
)

```

Output

?Enter your prompt: What is Artificial Intelligence

:Generated Response
 .Artificial Intelligence (AI) is the simulation of human intelligence by machines

Token	Log Prob	Probability	Cumulative	Color
Artificial	-0.02	0.9802	0.9802	Green
Intelligence	-0.03	0.9704	0.9512	Green
AI	-0.04	0.9608	0.9047	Green
simulation	-0.80	0.4493	0.2490	Red